ERGO TOOLS

Case Manual

VERSION 1.0

Acknowledgements

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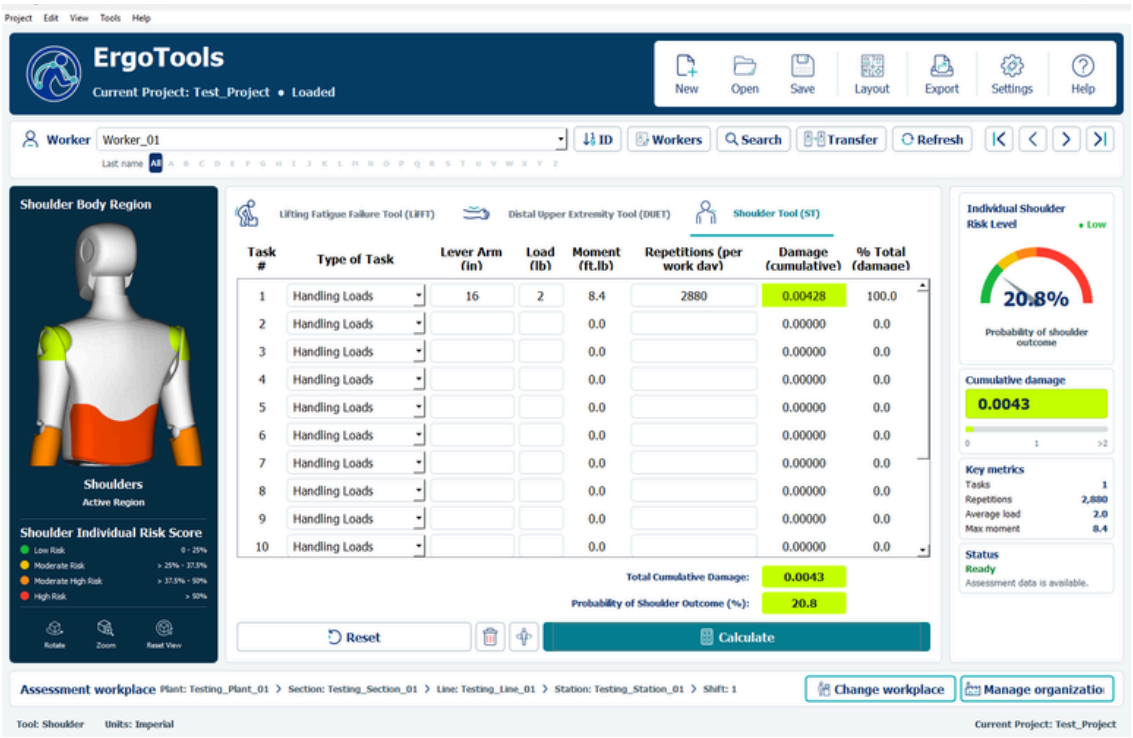
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CASE 1

Basic Setup & First Prompt



Objective: Create a new project, create a worker, assign worker and job information, enter analysis data, save the project, and reopen it successfully.

Preconditions

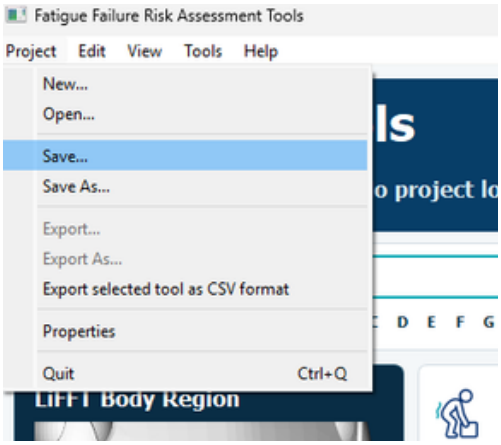
- The software shortcut is available on the desktop.
- A folder named “Projects” exists on the desktop.
- The user has access to the default test environment.

Steps

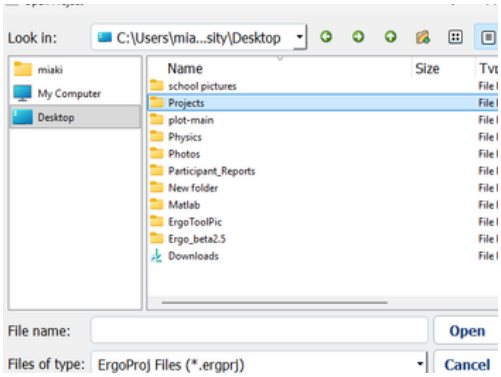
1. Launch the software
   - Double click **Run_ergotools – Shortcut** on the desktop.
   - If needed, drag the top bar of the software window to center it on the screen.

CASE 1

2. Create a project
    - Click “Project” in the upper-left corner.
    - Click “Save", or click the Save icon in the top-left toolbar.
    - In the save window, select Desktop from the left panel.
    - Open the “Projects” folder.
    - Enter the file name **Test_Project_01** and click “Save”
    - A pop-up window will request two inputs:
        - Name: **Test_Project**
        - Description: *This is a testing scenario*
    - Confirm the inputs



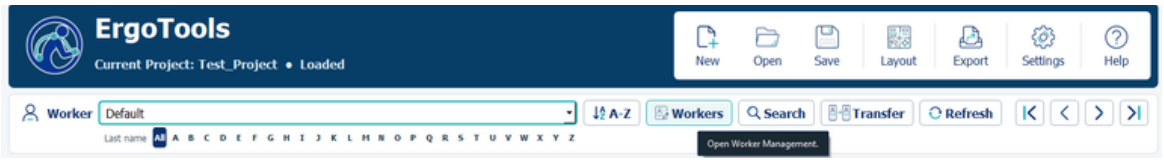




*__Expected result__: A confirmation message appears indicating that the project was saved successfully.



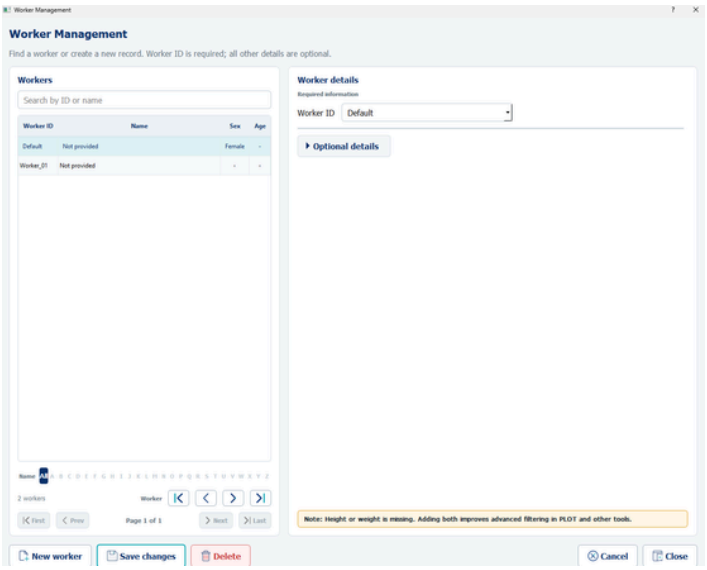
3. Create a worker
    - Click the Workers tab (second icon from left of the worker bar).



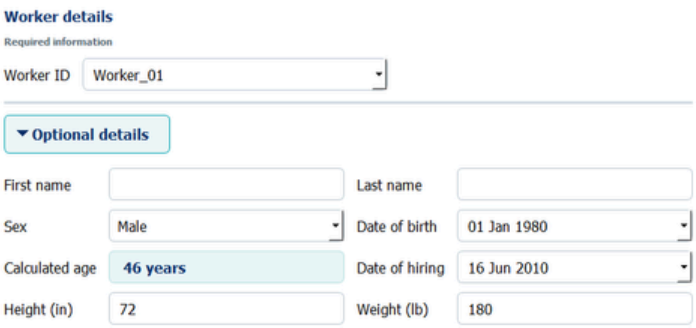
    - In the Workers Management window, click “New worker”.
    - Enter the Worker ID: **Worker_01**
    - Click “Save changes”
    - Confirm that the software displays a successful save message.

CASE 1

4. Remove the default worker entry
    - Click the button to return to Default.
    - Click "Delete"
    - Confirm the warning message.
    - Verify that **Worker_01** now becomes the active worker
    - Click "Close".

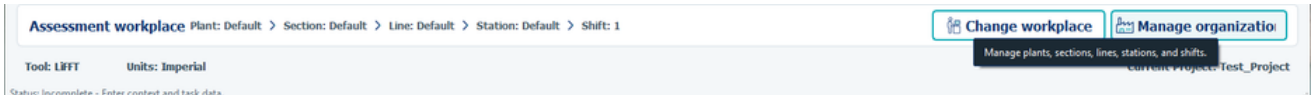


5. Enter personal characteristics
    - Reopen the worker editing window.
    - Click" Optional details".
    - Update the worker information as follows:
        - Date of Birth: *01/01/1980*
        - Date of Hiring*: 06/16/2010*
        - Sex: *Male*
        - Height: *72 inches*
        - Weight: *180*
    - Click "Save"
    - Close the worker management window.



*__Expected result:__ The worker record is saved with the updated demographic information.

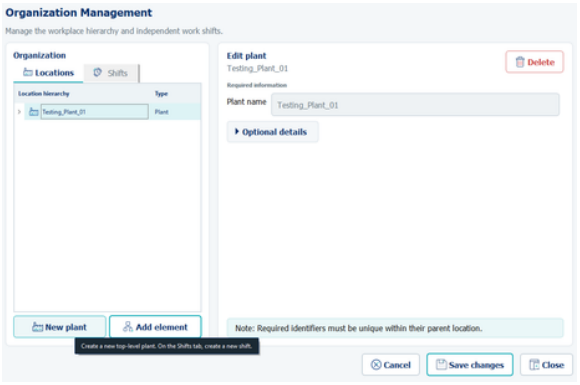
6. Manage organization
- On the main interface, located at the bottom of the screen, assign the worker to the following organizational levels:
    - Plant
    - Section
    - Line
    - Station
    - Shift



CASE 1

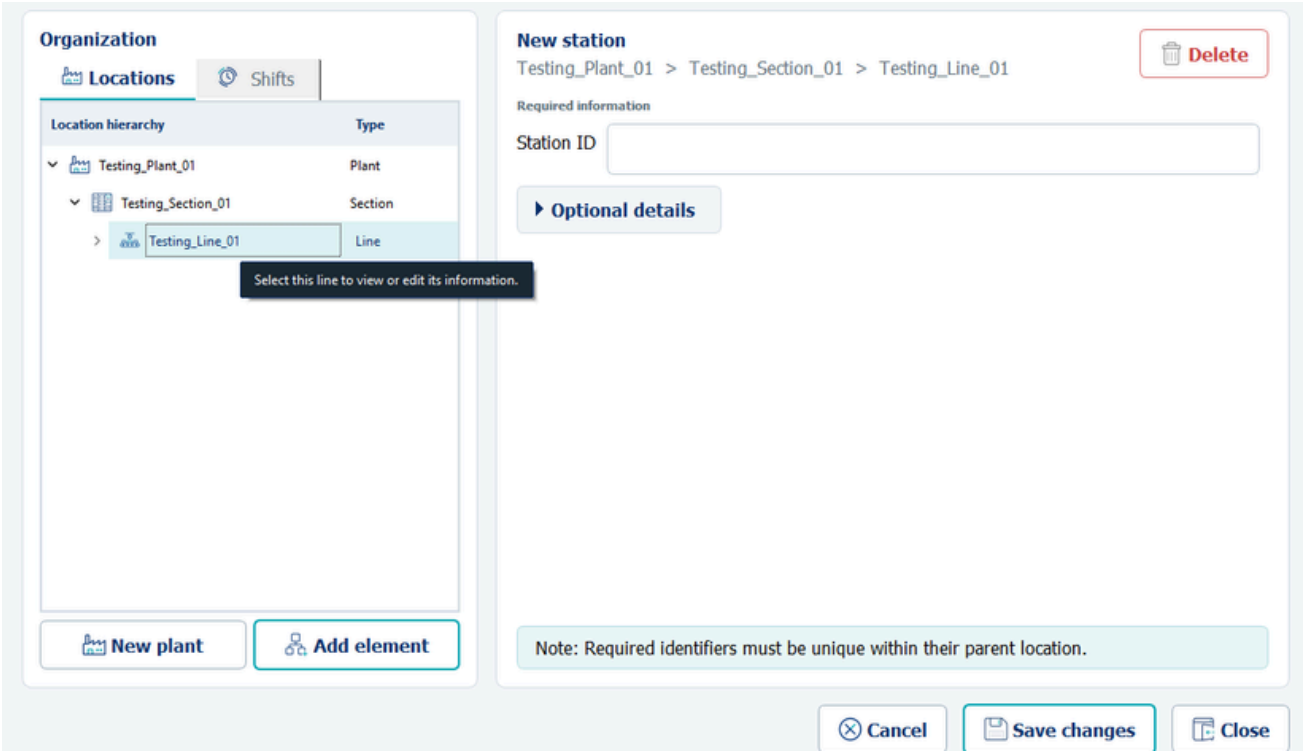
- For Plant:
  - In the Organization Management window, delete the "Default" entry first.
  - Click on the button "New Plant"
  - Enter:
    - Plant Name (ID): **Testing_Plant_01**
  - Click the button "Optional details"
  - Enter:
    - Description: *This is a testing plant*
    - Opening Date: *12/12/1970*
  - Click "Save"



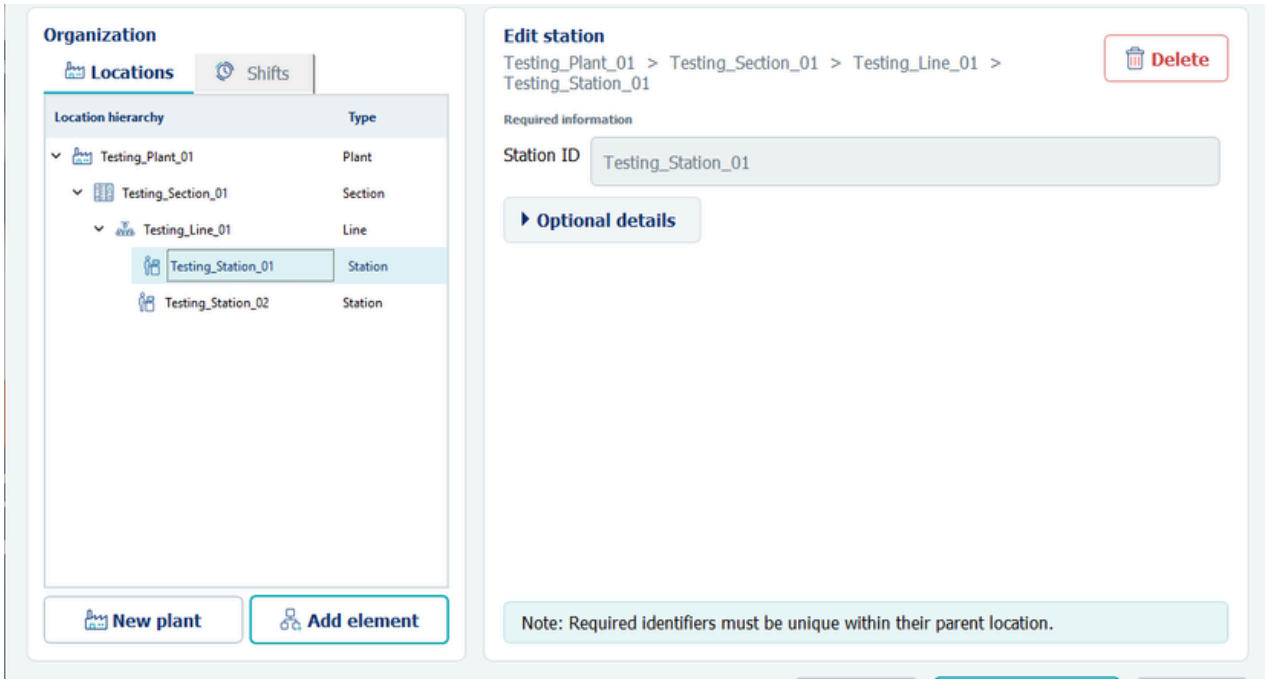


- For Section:
  - Click "Add element."
  - Enter **Testing_Section_01**
  - Click "Save"

- For Line:
  - Click "Add element."
  - Enter **Testing_Line_01**
  - Click "Save"

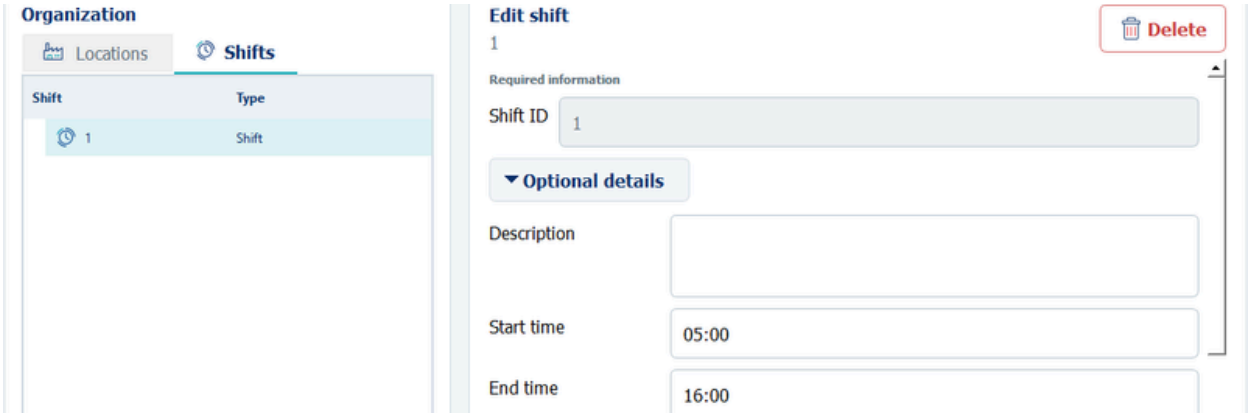


CASE 1



- For Station:
  - Click "Add element."
  - Enter **Testing_Station_01**
  - Click "Save"
  - Ensuring you are on **Testing_Line_01** of the "Location Hierarchy"
  - Click "Add element."
  - Enter **Testing_Station_02**
  - Click "Save"

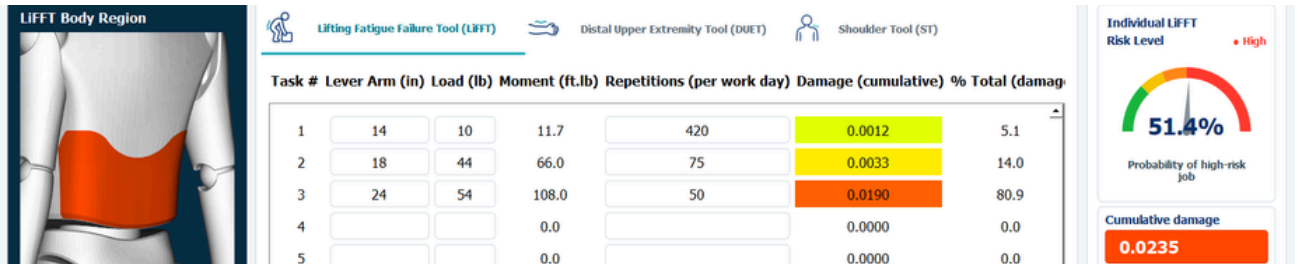
- For Shift:
- Open the "Shifts" tab
- Keep the existing Shift ID.
- Change the end time from 5:00 to 16:00
- Click "Save," then "Close."



CASE 1

7. Enter analysis data

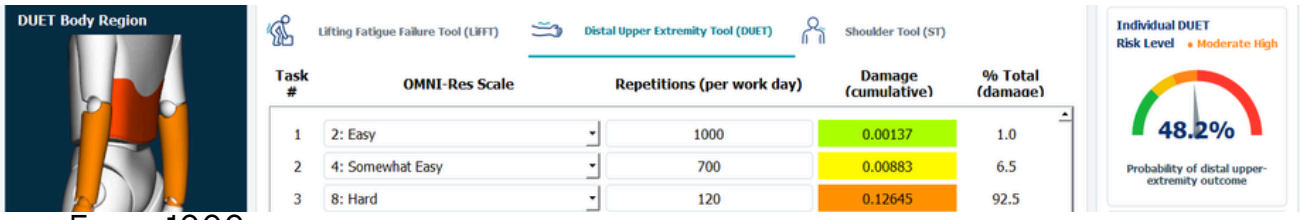
- In the LIFFT tab, enter the three tasks shown in the reference image.



14 10 420
18 44 75
24 54 50

  - Click "Calculate"

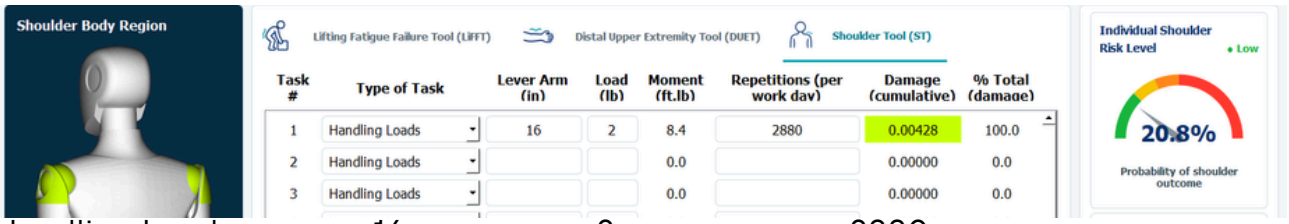
- Go to the DUET tab, enter the tasks shown in the reference image.



Easy : 1000
Somewhat Easy : 700
Hard : 120

  - Click "Calculate."

- Go to the ST tab, enter the three tasks shown in the reference image.



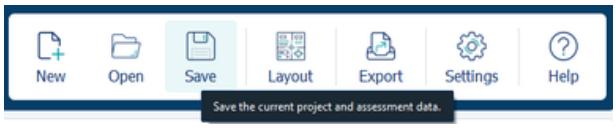
Handling Loads: 16 2 2880

  - Click "Calculate"

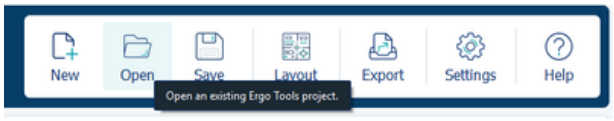
*__Expected result__: Each tool calculates successfully and displays results on screen.

CASE 1

8. Save and reopen the project
    - Click Save.
    - Confirm any save notifications.
    - Close the software.

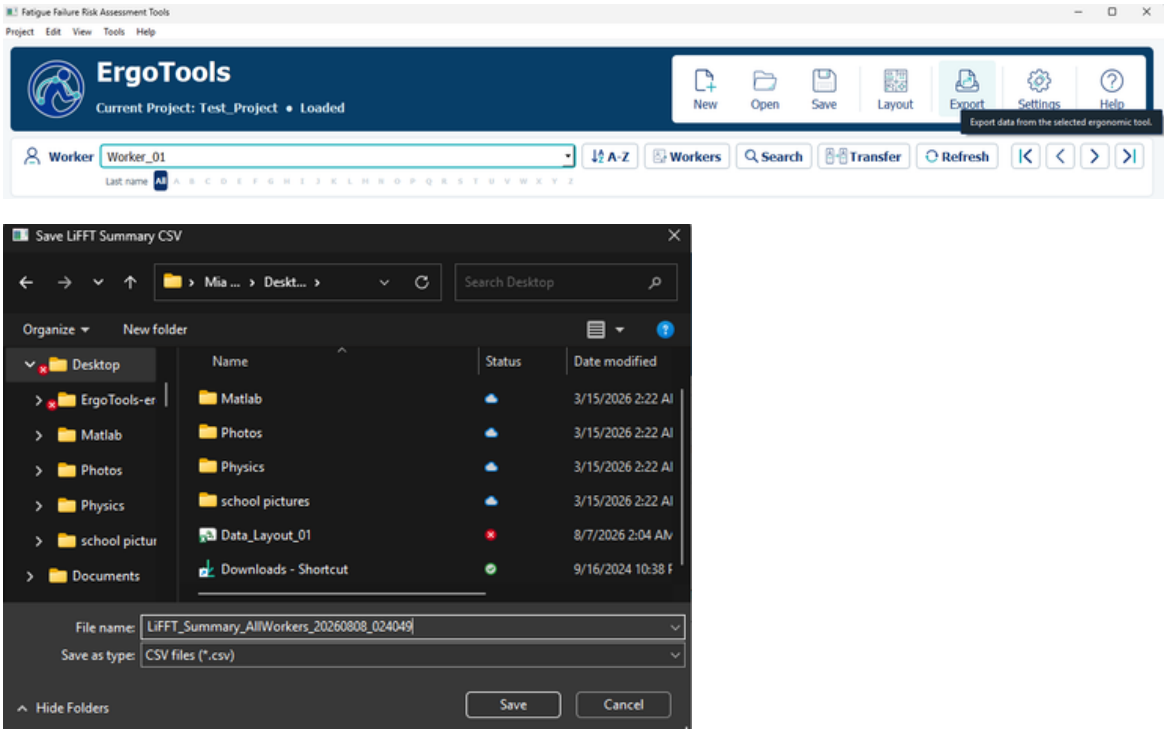


    - Reopen the software.
    - Click Open (folder icon).
    - Open the previously saved project.



***Expected result:** The project reopens with the previously entered worker, job assignments, and analysis data loaded correctly.

9. Export the file to CSV
    - Click on "Export" at the top

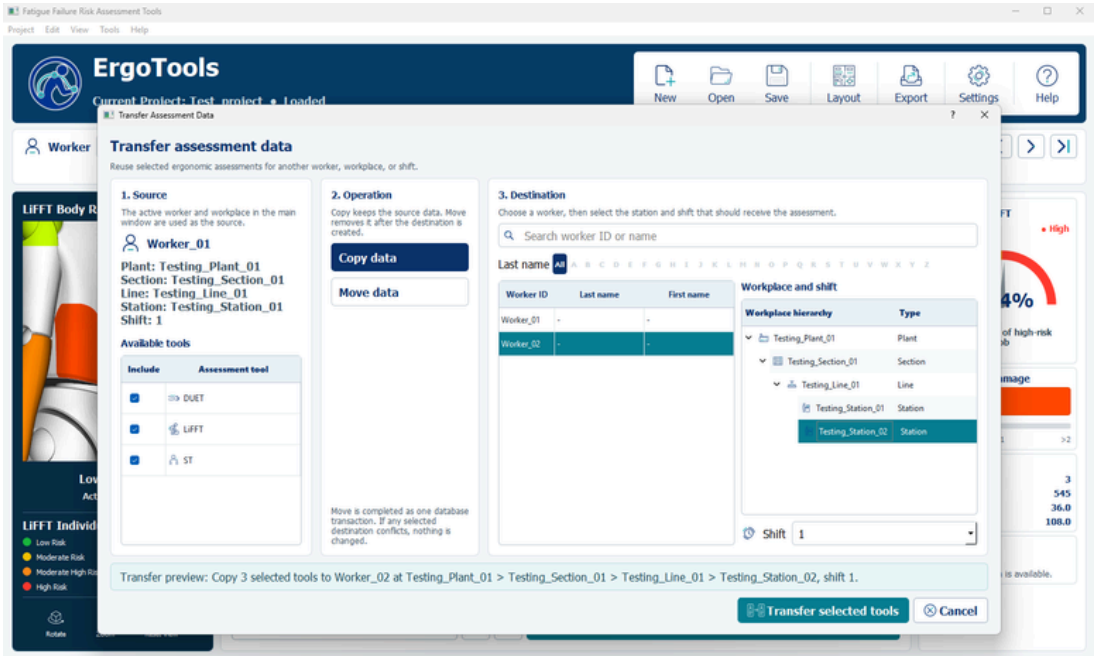


    - Find a location to save your file, it will have the selected tool results for all workers in the project.

***Expected result:** The user will have a csv file with the results of the analyses.

CASE 2

Project Management



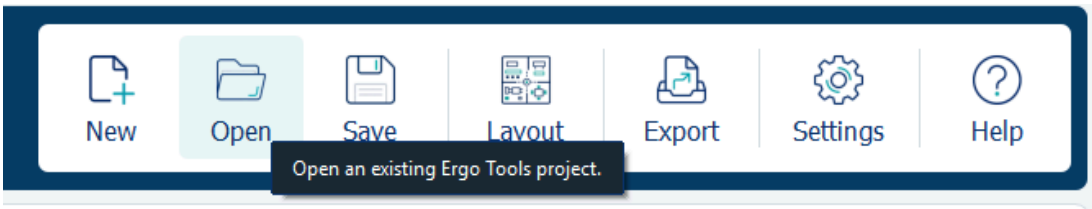
Objective: Duplicate existing analyses from one worker to another worker who works in the same plant, section, and line, but at a different station.

Preconditions

- The project from Case 1 already exists.
- **Worker_01** contains saved analyses.

Steps

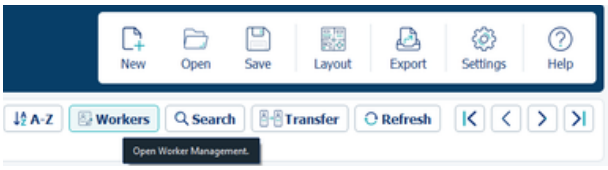
1. Launch the software
   - Open the existing project
   - Open the project created in Case 1.

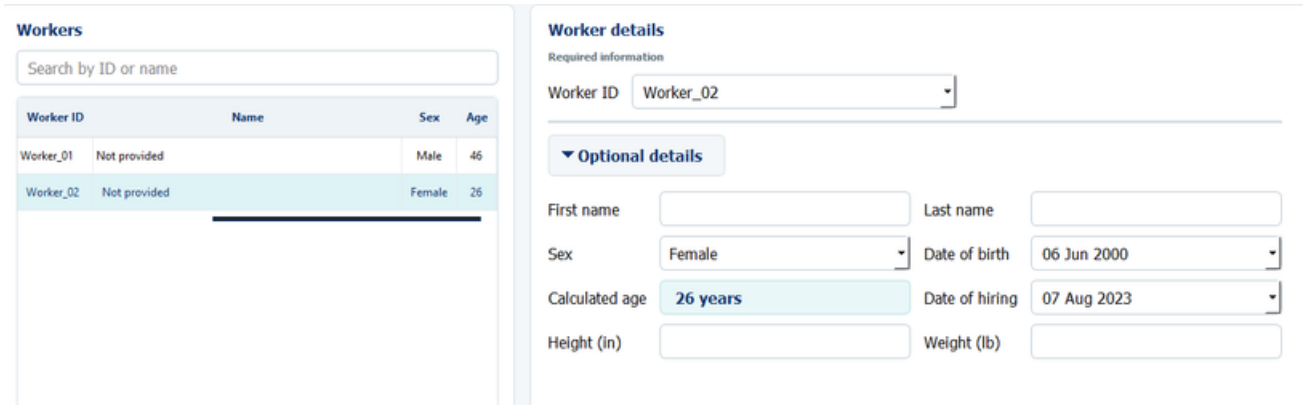


   - Confirm that **Worker_01** and the previously entered data are loaded.

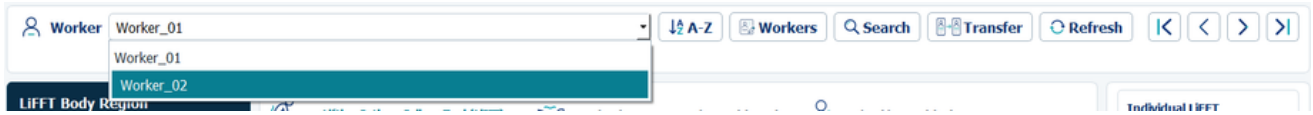
CASE 2

2. Create a second worker
    - Open the worker editing window.
    - Click “New”.
        - Enter the Worker ID: **Worker_02**
        - Date of Birth: *06/06/2000*
        - Date of Hiring*: 08/07/2023*
        - Sex: *Female*
    - Click Save.
    - Close the worker window.





3. Switch to the new worker
    - On the main interface, use the worker drop-down list.



    - Select **Worker_02**.

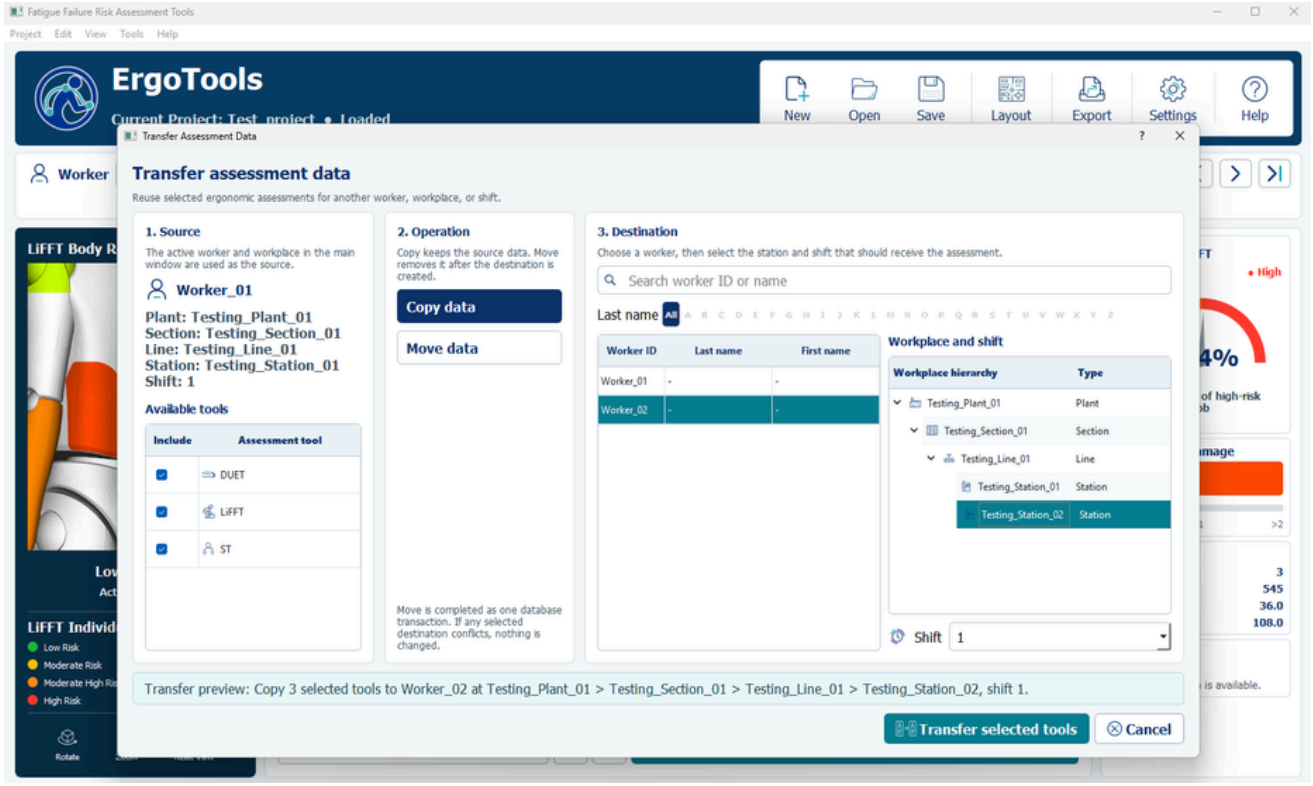
Expected result:* **Worker_02 is displayed on the main screen, but the analysis tabs are empty.

4. Transfer data from **Worker_01**
    - Switch back to **Worker_01.**
    - Click the Transfer Data button.



CASE 2

- In the Transfer Worker Tool Data window:



- Select **Worker_02** as the destination worker.
- Change the destination station to **Testing_Station_02**.
- Keep “Copy Data” selected.
- Make sure that all available tools are selected.
- Click “Transfer selected tools”.
- Confirm the transfer for each tool when prompted.
- Close the transfer window.

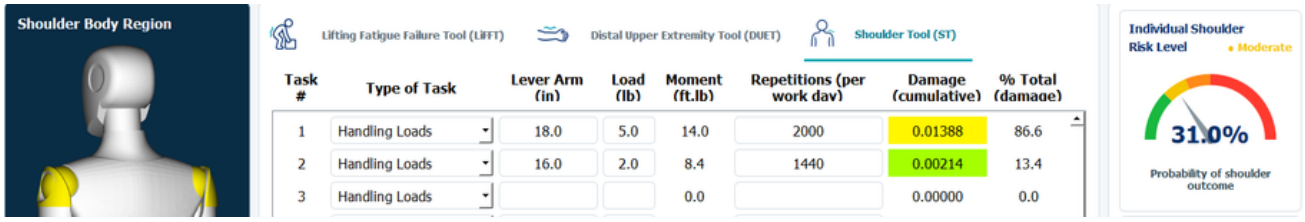
***Expected result:** The analyses from Worker_01 are copied to Worker_02 under Testing_Station_02.

CASE 2

5. Verify transferred data
    - Select **Worker_02** on the main interface.

*If no analysis appears immediately, change the station to **Testing_Station_02**.

    - Open the “Shoulder Tool” tab.
    - Update the input values according to the reference image.

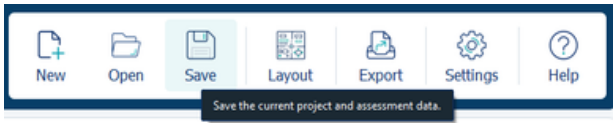


Handling Loads: 18 5 2000
Handling Loads: 16 2 1440

    - Click “Calculate”

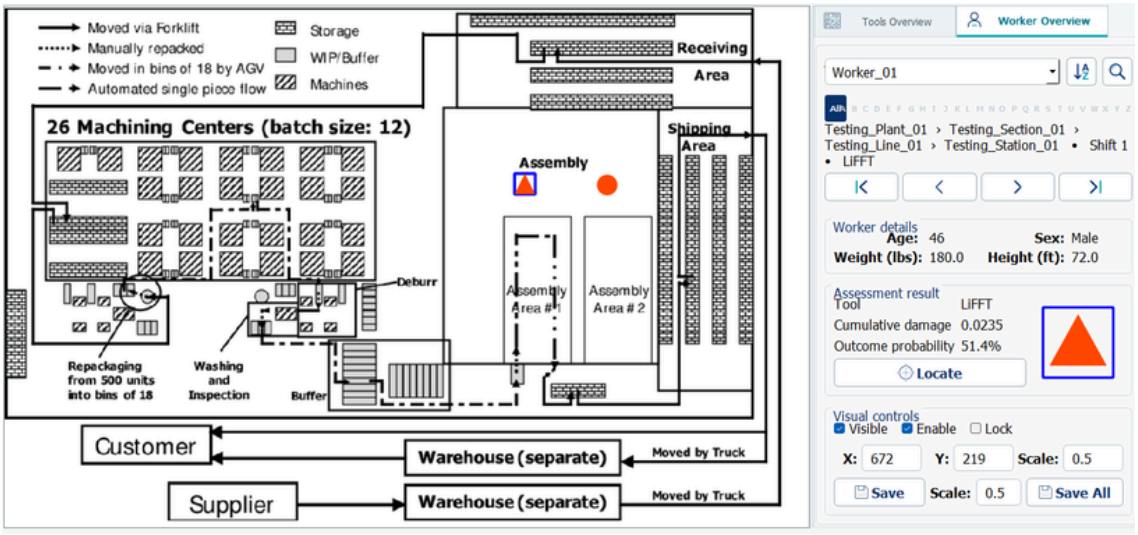
6. Save the project
    - Click the “Save” button in the top toolbar.
    - Confirm the changes.

*__*Expected result:*__ The project is saved with both workers and their corresponding analyses.



CASE 3

Layout View



Objective: Open the layout view and load a predefined layout.

Preconditions

- The project from Case 2 is already open.
- The predefined layout file exists in the assets folder.
- At least two workers are available in the project.
- Worker assignments have already been saved in the project.

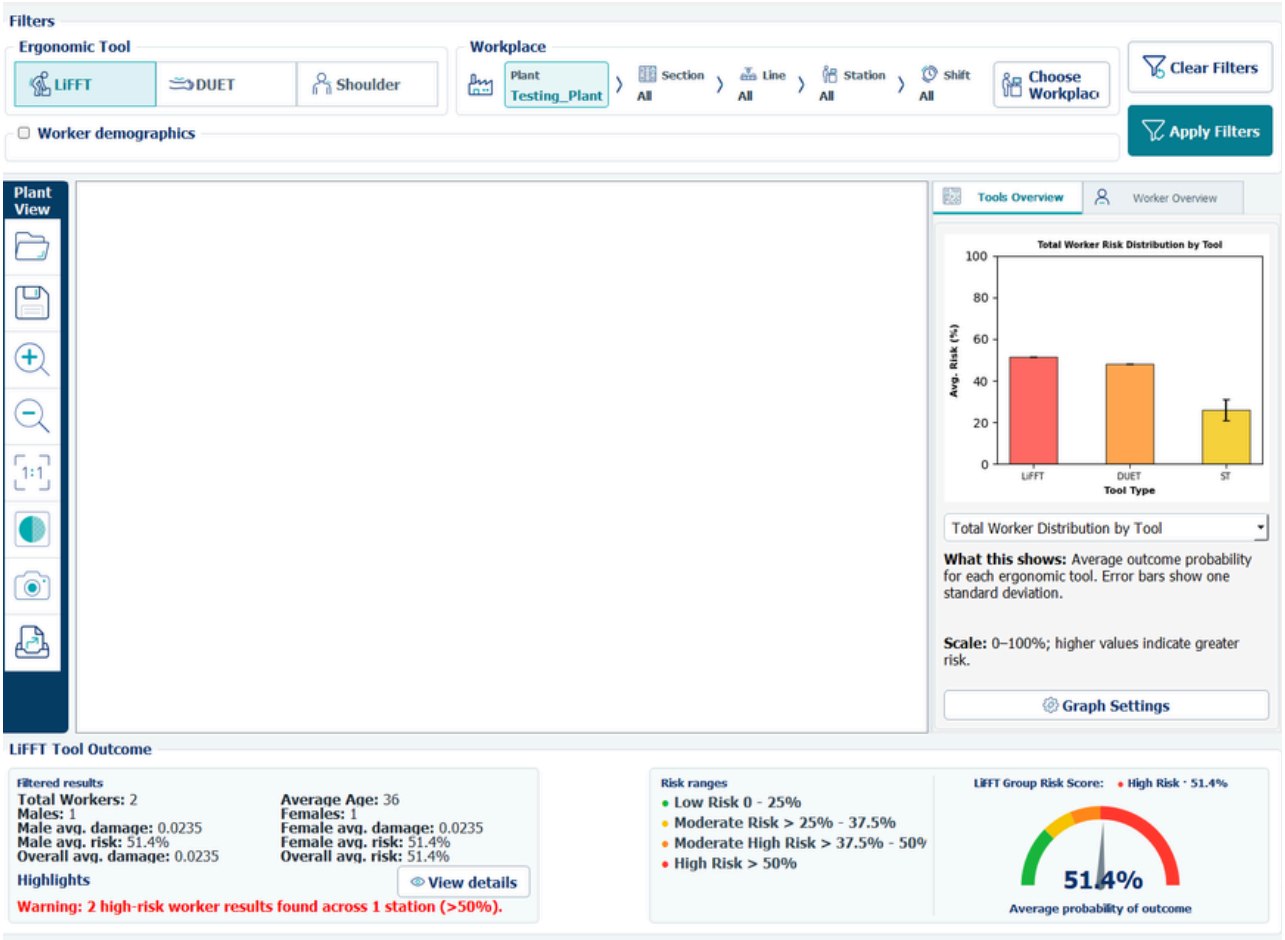
Steps

1. Open Layout View
   - With the project open, click Layout to enter PLOT's Layout View.

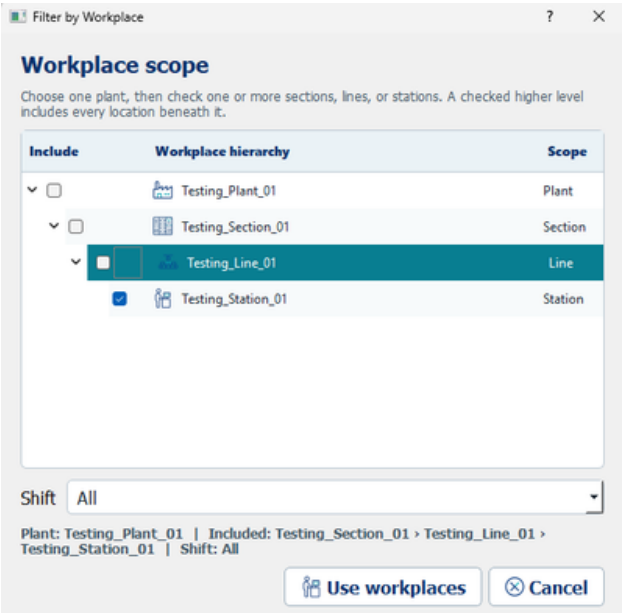


   - Confirm that the Layout View window opens successfully. The following window will appear:

CASE 3



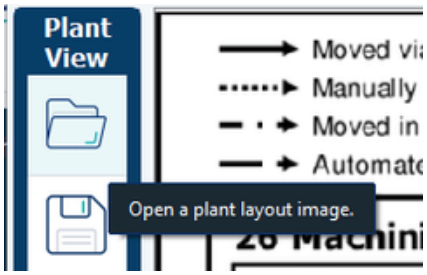
- At the top of the window, identify the filtering sections for Plant, Section, Line, Station, and Shift.
- Click on "Choose Workplace"
- Click on the box by "Testing_Station_01"
- Click on "Use workplaces"



Expected result: The Layout View opens and the user can identify the filtering options and tool-selection panel.

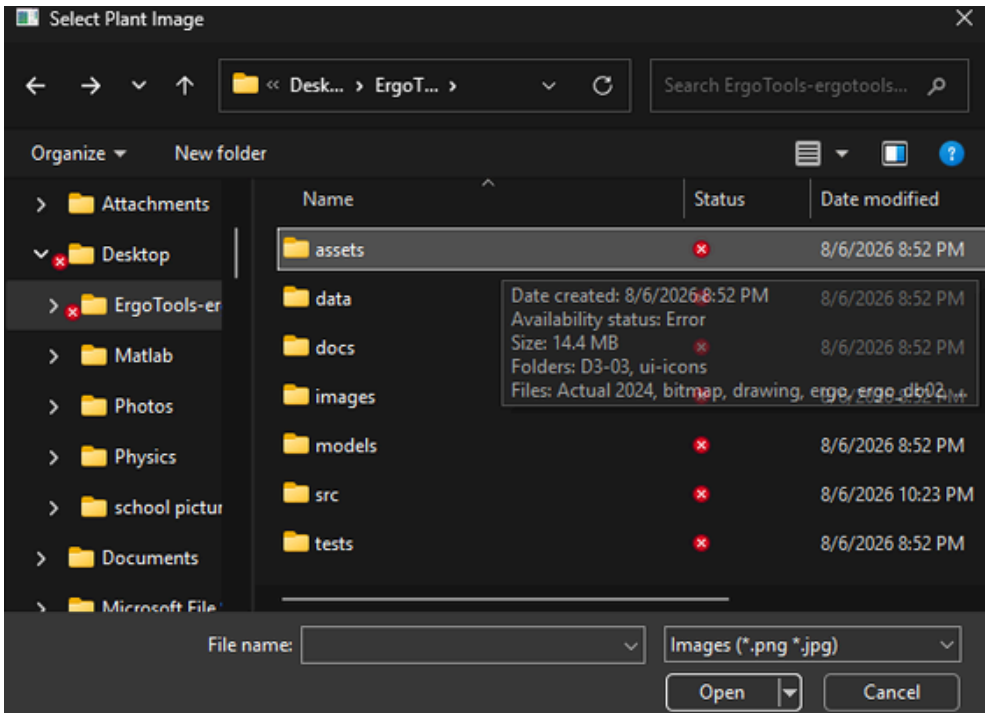
CASE 3

2. Load a predefined layout
   - Click the Open Layout icon on the left side of the Layout View.

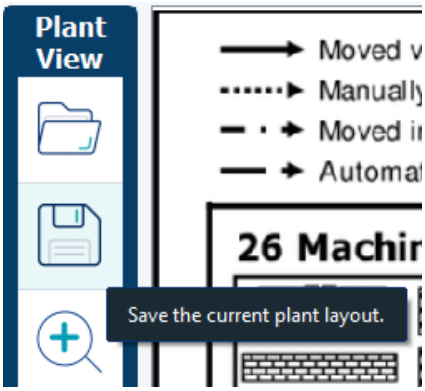


   - Browse to:
   - **C:\Users\user01\Desktop\ErgoTools-plot\assets**

*If your Windows username is different, update the folder path accordingly.

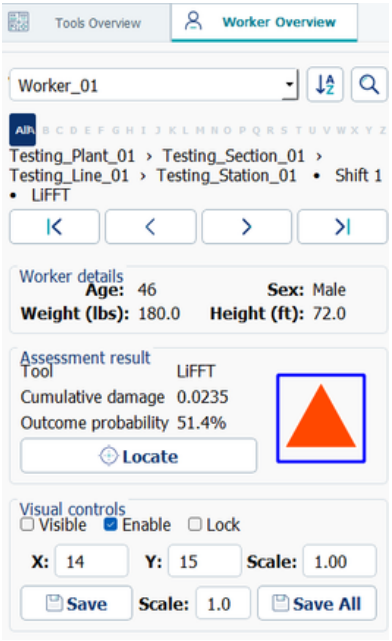


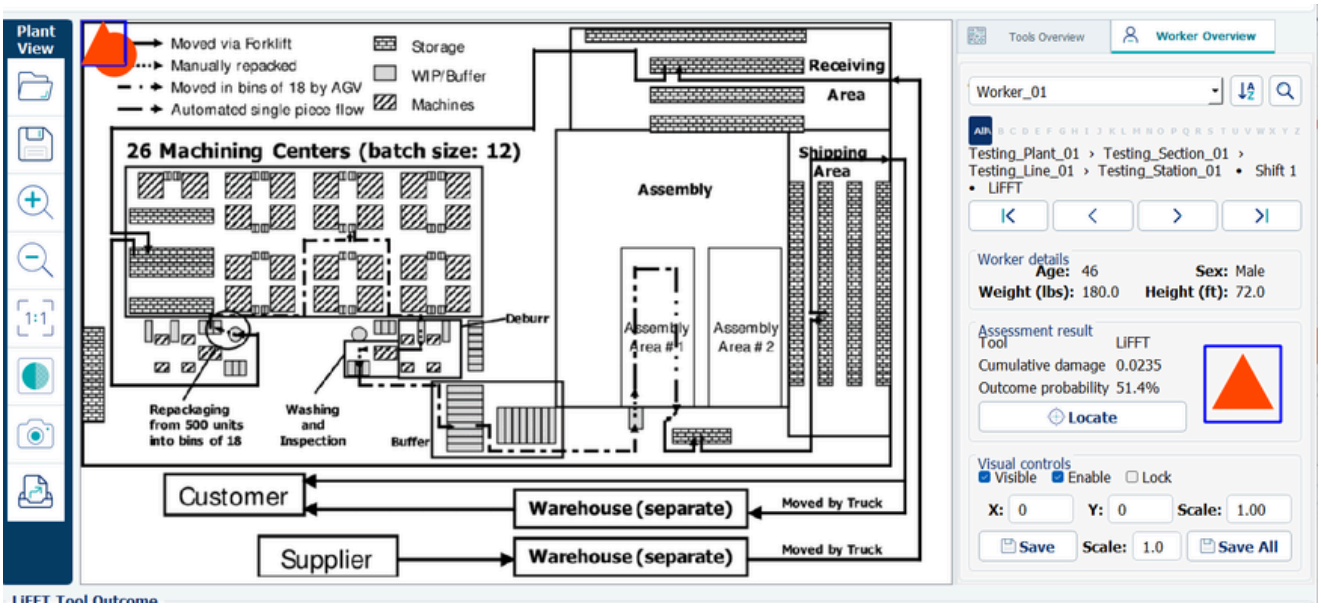
   - Select **layout01**.
   - Click "Open".
   - Back in the Layout View, click "Save" to save the loaded layout.
   - Confirm that a notification appears indicating that the plant image has been saved successfully.



CASE 3

3. Display workers in the layout
    - Confirm that no workers appear initially in the layout.
    - To the right of the “Tools Overview”, locate the “Worker Overview.”
    - In the “Worker ID” drop-down list, select **Worker_01**.
    - Check the Visible option to make the worker appear in the layout
    - Confirm that the worker appears as a geometric figure, with the shape based on sex and the color based on the currently selected analysis tool (Circle for female, triangle for male).



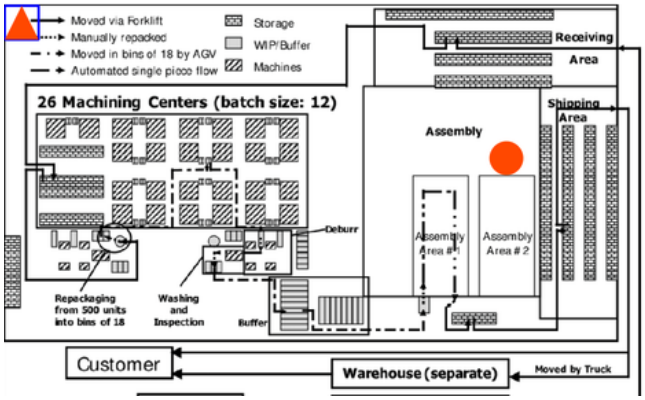


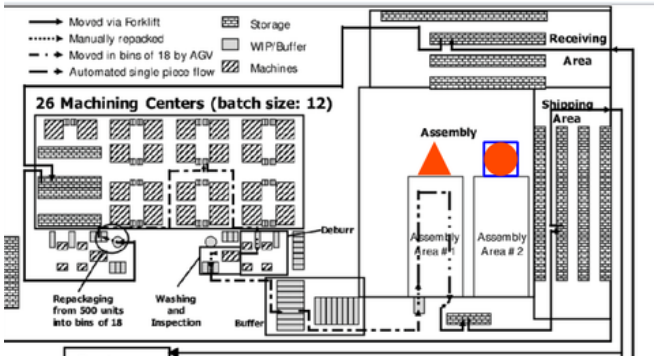
    - Select **Worker_02** from the Worker ID drop-down list.
    - Check the Visible option for **Worker_02**.
    - Confirm that the second worker also appears in the layout.

Expected result:* The selected workers become visible in the layout.

CASE 3

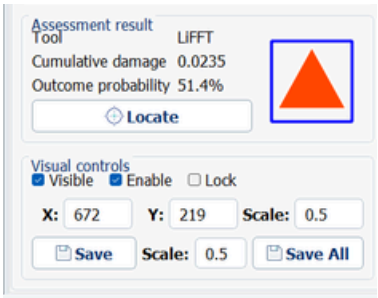
4. Reposition workers in the layout
    - Click and hold the figure for **Worker_02** (Circle).
    - Drag the figure to the desired location, for example just below the Assembly section shown in the reference image.
    - Click and hold the figure for **Worker_01** (Triangle)
    - Drag the figure to the desired location, for example next to **Worker_02**



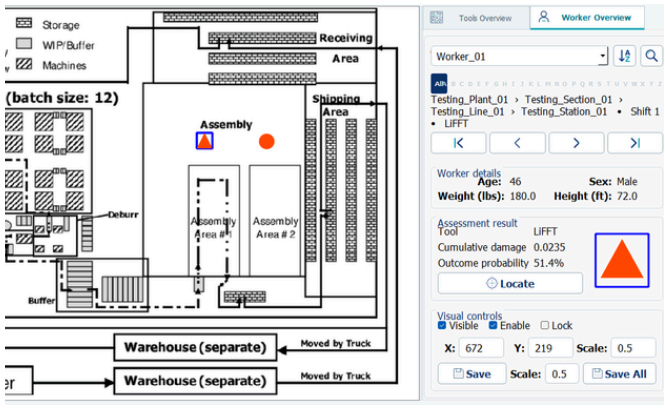


*__Expected result__: Both worker icons can be moved to user-defined positions within the layout

5. Adjust worker icon scale
    - In the bottom section of the "Worker Overview", locate the Scale options.
    - The "Scale" on the X Y line is for individual workers
    - The "Scale" at the bottom is for all workers
    - Change the "Scale" for all workers to 0.5.
    - Click "Save All"
    - Confirm that the worker icons are resized and that a confirmation message appears.



*__Expected results:__ The visible worker icons are resized successfully.



6. Save and close the layout
    - Confirm that the layout shows both workers in their selected positions.
    - Close the Layout View window.
    - Save the project.

CASE 4

Large Project Navigation, Layout Review, and Export



Objective: Load a project with a large worker database, navigate the layout using filters and tools, inspect individual workers from the layout, and export the layout image and displayed data successfully.

Preconditions

- The Case 4 folder is available and contains the required project files.
- The file **Case_04.ergprj** is available inside the Case 4 folder.
- The database is already populated with workers and analysis results.
- The software shortcut is available on the desktop.

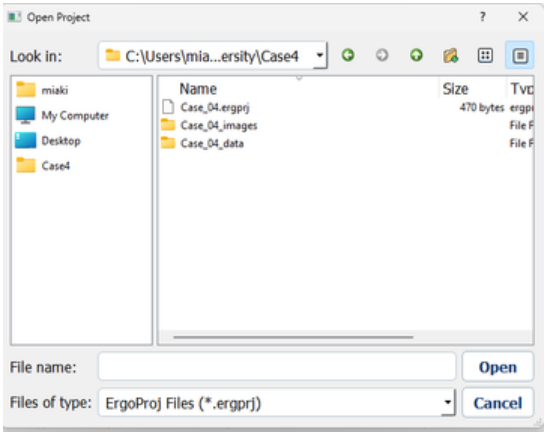
Steps

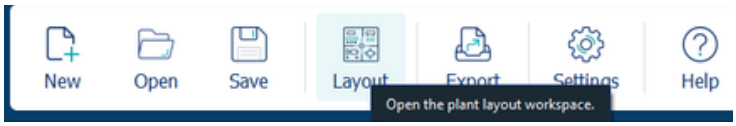
1. Launch the software
    - Double-click **Run_ergotools – Shortcut** on the desktop
    - If needed, drag the top bar of the software window to center it on the screen.

**Expected result:* The software opens successfully.

CASE 4

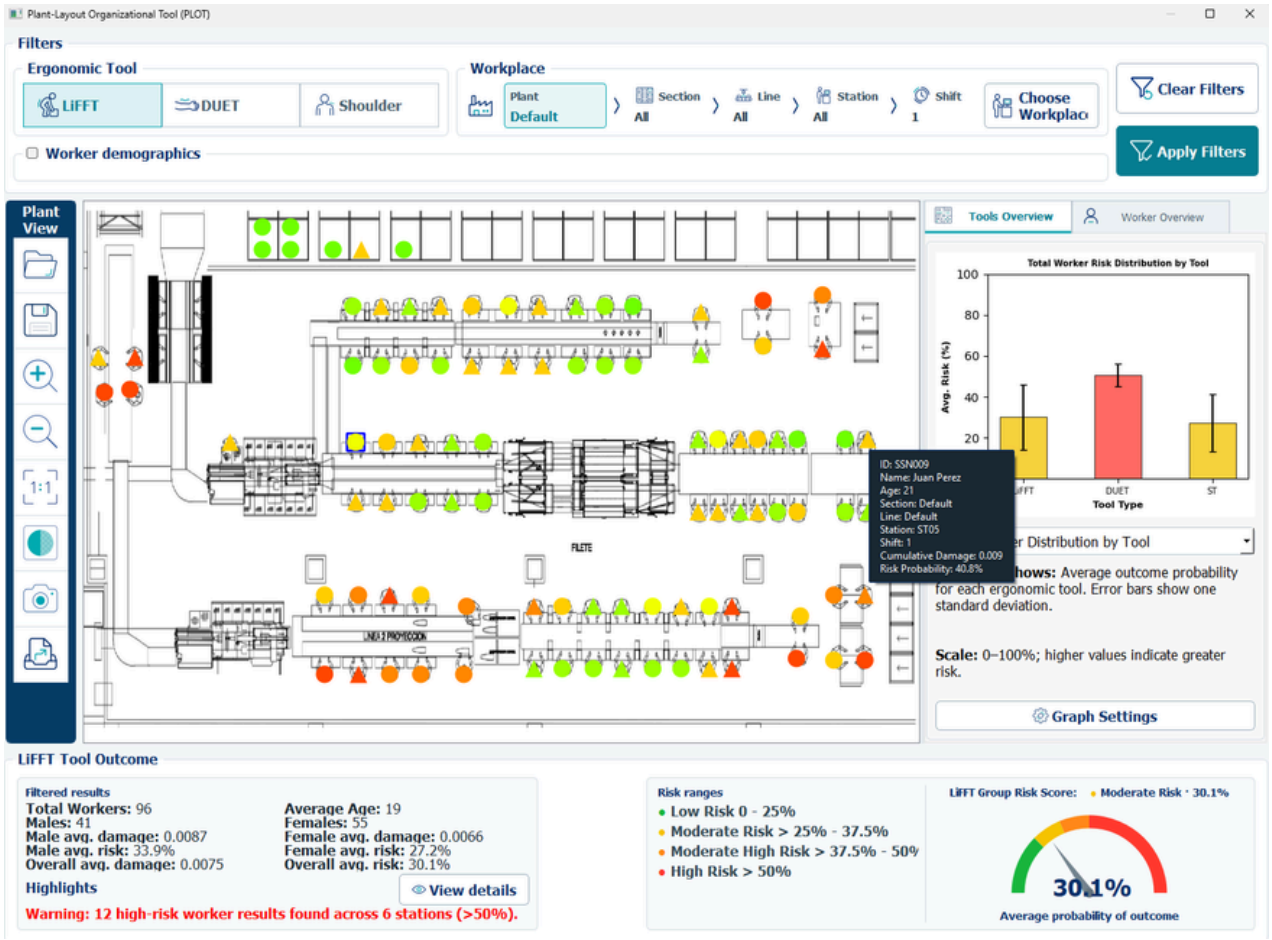
2. Load a project
    - Click Project in the upper-left corner.
    - Click Open, and select Case 4 folder, inside open the file **Case_04.ergprj**.
    - Click Open.
    - Confirm that the project loads successfully.
    - Click Layout to open the Layout View.



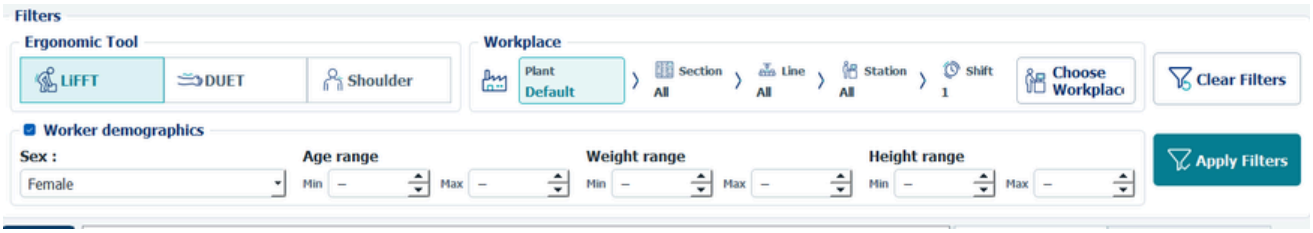


***Expected result:** The populated project opens successfully and the Layout View can be accessed.

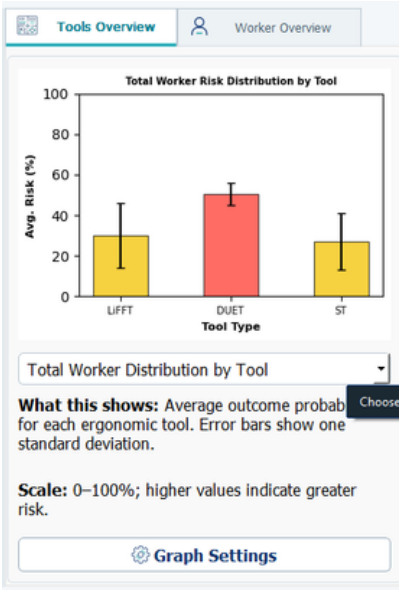
3. Review the Layout View interface
    - Confirm that the Layout View opens successfully.



CASE 4



- Identify the filtering sections available in the layout.
- Identify the tool-selection area used to switch between analysis views, such as LIFFT and DUET.
- Confirm that worker information and plots are visible in the populated layout environment.



*__Expected result:__ The user can identify the main layout controls, filters, plots, and worker visualization area.

4. Navigate between analysis tools
   - Select the LIFFT view in the Layout View.
   - Observe the displayed worker results and plots.
   - Switch to the DUET view.
   - Apply the desired filter.



Layout with DUET filter

   - Switch back to LIFFT.
   - Apply the filter again.

*__Expected result:__ The layout updates when different analysis tools are selected, and the displayed worker information reflects the selected tool.

CASE 4

5. Apply and clear worker filters
    - In the "Worker demographics" section locate the Sex filter.
    - Change the filter from Both to Female.
    - Click "Apply Filters"
    - Confirm that only female workers remain displayed.

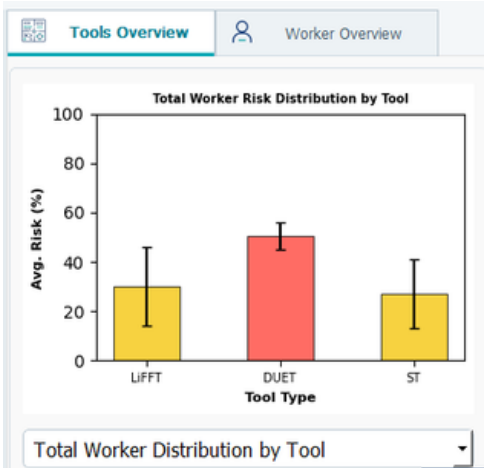




    - Change the filter back from Female to Both.
    - Click "Apply Filters"
    - Confirm that the full worker population is displayed again

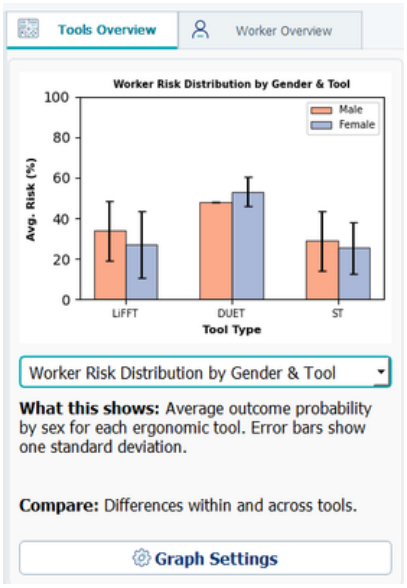
*__Expected result:__ The layout correctly updates when the worker filter is changed and reapplied.

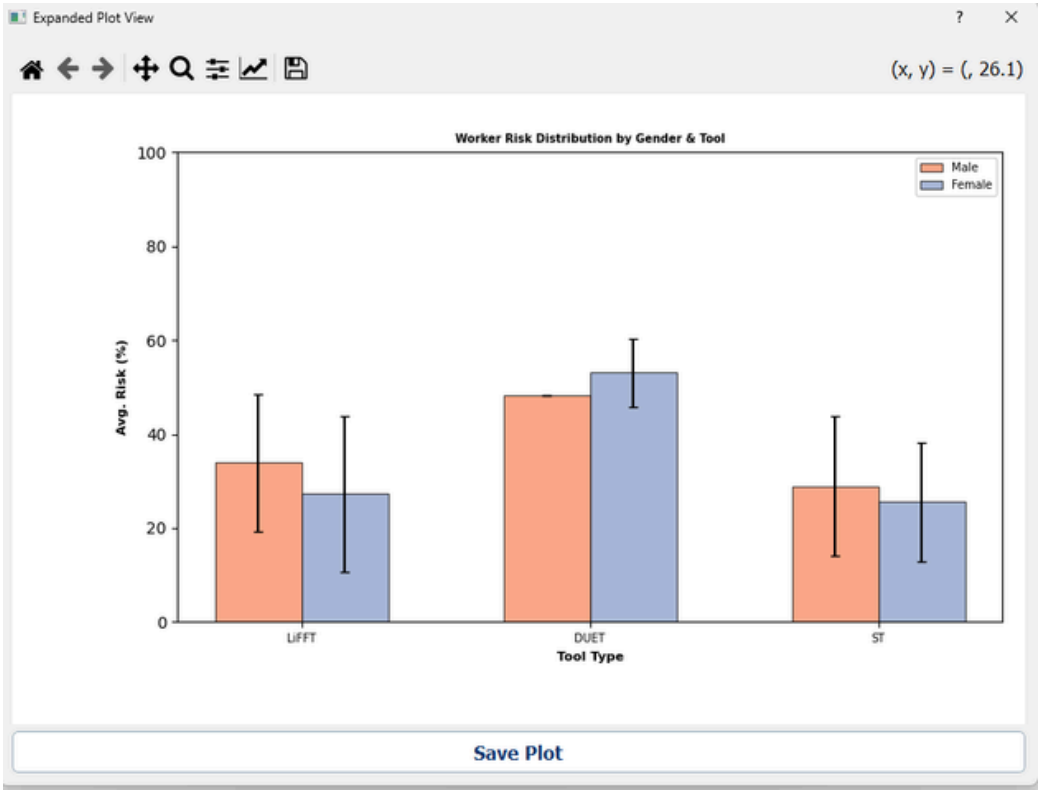
6. Review layout plots
    - Locate the plot showing "Total Worker Distribution by Tool."



CASE 4

- Change the displayed plot to the "Worker Risk Distribution by Gender & Tool."
- The plot will change without having to click on "Apply Filters"
- Click the plot to open it in a larger window.
- Review the enlarged plot.
- Save the enlarged plot as an image by selecting a folder, use as file name **Testing_Risk_By_Gender_And_Tool**.
- Close the enlarged plot window.
- Return to the original plot view for total worker distribution by tool.





Expected result: The selected plot opens in a larger window and can be saved as an image.

CASE 4

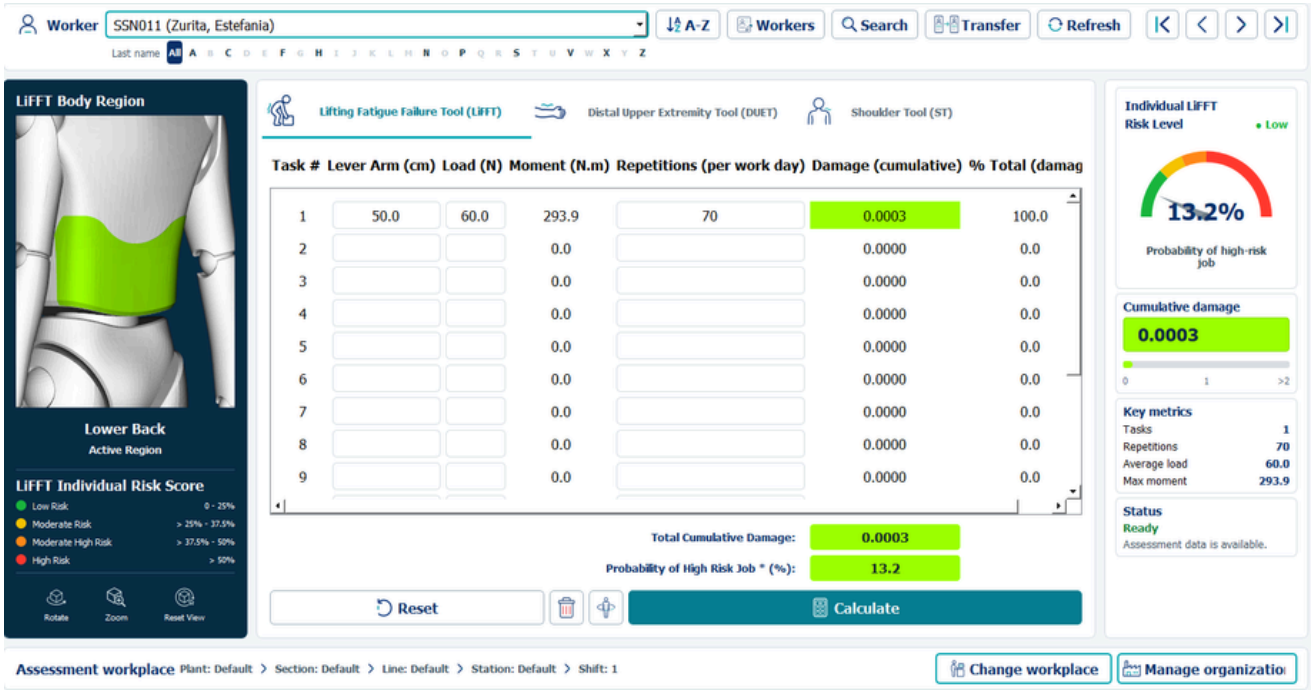
7. Inspect an individual worker from the layout
    - In the Layout View, hover your mouse over one worker icon.
    - Confirm that worker information appears



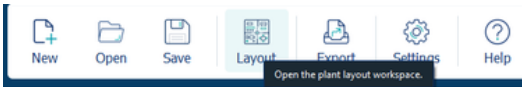
    - Right-click on the same worker icon.
    - Select “Load into Tool Window.”
    - Confirm that the Layout View closes and the software returns to the main interface.
    - Confirm that the selected worker and that worker’s analyses are loaded into the tool window.

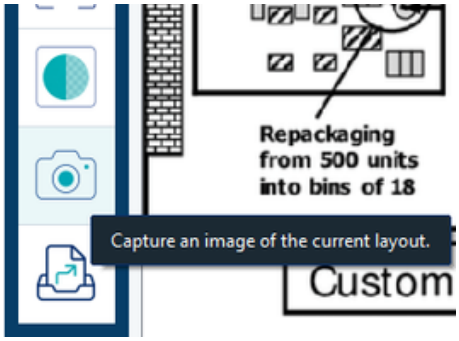
*__Expected result:__ The selected worker can be loaded directly from the Layout View into the main tool window.

CASE 4



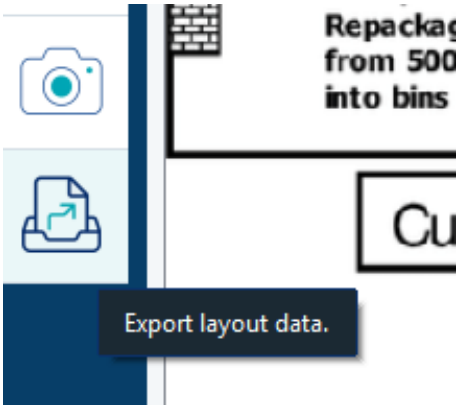
8. Reopen the layout and export the layout image
    - Return to Layout View.
    - Click the Camera icon on the left side of the layout.
    - In the save window, choose the Desktop as the destination folder.
    - Enter the file name **Testing_Layout_Image**.
    - Save the file.
    - Confirm that the layout is exported as a PNG image file.





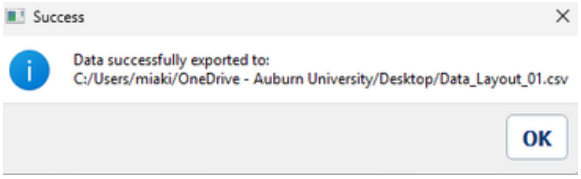
*__*Expected result:*__ A PNG image of the layout is saved successfully.

9. Export the displayed layout data
    - Click the Export Data icon on the left side of the layout.
    - In the save window, choose the Desktop as the destination folder.



CASE 4

- Enter the file name *Data_Layout_01.*
- Save the file.
- A confirmation message will appear.
- Confirm the message, by clicking “OK”



***Expected result:** A CSV file containing the displayed layout data is saved successfully.

10. Verify exported files

- Open the saved **Layout_01.png** file from the Desktop.
- Confirm that the image contains the expected layout view.
- Open the saved **Data_Layout_01.csv** file from the Desktop.
- Confirm that the file contains the expected exported layout data.

***Expected result:** Both export files are created successfully and contain the expected information.

